Lay Lec 5

L5_A_S2.pdf
PDF

> "Generate a lecture scribe intended to serve as reference material for exam preparation. Use *ONLY* the provided context (Lecture 5 slides).
> *Instructions:*
> 1. *Purpose:* The scribe must be exam-ready, strictly following the content of the slides.
> 2. *Content:* Include definitions, assumptions, theorem statements, proofs or derivations, and worked examples *exactly* as presented in the lecture materials.
> 3. *Constraints:* Do not introduce new examples, outside information, or hallucinations. Do not use phrases like 'for example' to add your own content.
> 4. *Structure:* Organize the scribe clearly with appropriate headings corresponding to the lecture flow.
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> *Goal:* Produce a clean, accurate revision document based solely on the attached file."

Below is a **clean, exam-ready lecture scribe** prepared **strictly from the provided Lecture 5 slides**.
No external material, interpretations, or additional examples have been added.
All notation, definitions, statements, and worked examples follow the **lecture flow and wording** as presented.

CSE400 – Fundamentals of Probability in Computing

Lecture 5: Bayes' Theorem, Random Variables, and Probability Mass Function

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1. Bayes' Theorem

1.1 Weighted Average of Conditional Probabilities

Let A and B be events.

We may express event A as:

$$A = AB \cup AB^c$$

This follows because, for an outcome to be in A, it must either:

- be in both A and B, or
- be in A but not in B.

Since AB and AB^c are mutually exclusive, by **Axiom 3**:

$$\Pr(A) = \Pr(AB) + \Pr(AB^c)$$

Using conditional probability:

$$\Pr(A) = \Pr(A \mid B)\Pr(B) + \Pr(A \mid B^c)[1 - \Pr(B)]$$

Interpretation:
The probability of event A is a **weighted average of conditional probabilities**, where the weights are the probabilities of the events on which the conditioning is done.

1.2 Learning by Example

Example #3.1 (Part 1/2)
An insurance company divides people into two classes:

- accident prone
- not accident prone

Given:

- $\Pr(\text{accident} \mid \text{accident prone}) = 0.4$
- $\Pr(\text{accident} \mid \text{not accident prone}) = 0.2$
- $\Pr(\text{accident prone}) = 0.3$

Question:
What is the probability that a new policyholder will have an accident within one year?

Solution:

Let:

- A_1: policyholder has an accident within one year
- A: policyholder is accident prone

$$\Pr(A_1) = \Pr(A_1 \mid A)\Pr(A) + \Pr(A_1 \mid A^c)\Pr(A^c)$$

$$= (0.4)(0.3) + (0.2)(0.7) = 0.26$$

Example #3.1 (Part 2/2)
Question:
Suppose a new policyholder has an accident within a year. What is the probability that they are accident prone?

Solution:

$$\Pr(A \mid A_1) = \frac{\Pr(AA_1)}{\Pr(A_1)}$$

$$= \frac{\Pr(A)\Pr(A_1 \mid A)}{\Pr(A_1)} = \frac{(0.3)(0.4)}{0.26} = \frac{6}{13}$$

1.3 Formal Introduction: Law of Total Probability

Suppose $B_1, B_2, \ldots, B_n$ are mutually exclusive events such that:

$$\bigcup_{i=1}^{n} B_i = B$$

Exactly one of the events $B_1, B_2, \ldots, B_n$ must occur.

Writing:

$$A = \bigcup_{i=1}^{n} AB_i$$

Since the events AB_i are mutually exclusive:

$$\Pr(A) = \sum_{i=1}^{n} \Pr(AB_i) = \sum_{i=1}^{n} \Pr(A \mid B_i) \Pr(B_i)$$

This is called the **Law of Total Probability (Formula 3.4)**.

1.4 Bayes Formula

Using:

$$\Pr(AB_i) = \Pr(B_i \mid A) \Pr(A)$$

We obtain:

$$\Pr(B_i \mid A) = \frac{\Pr(A \mid B_i) \Pr(B_i)}{\sum_{j=1}^{n} \Pr(A \mid B_j) \Pr(B_j)}$$

This is known as the **Bayes Formula (Proposition 3.1)**.

- $\Pr(B_i)$: **apriori probability**
- $\Pr(B_i \mid A)$: **posteriori probability**

1.5 Learning by Example

Example #3.2 (Card Problem)
Three cards:

- one red–red (RR)
- one black–black (BB)
- one red–black (RB)

A card is chosen at random and placed on the ground.
The upper side is red.

Question:
What is the probability that the other side is black?

Solution:

Let:

- RR, BB, RB denote the chosen card type
- R: upturned side is red

$$\Pr(RB \mid R) = \frac{\Pr(R \mid RB)\Pr(RB)}{\Pr(R \mid RR)\Pr(RR) + \Pr(R \mid RB)\Pr(RB) + \Pr(R \mid BB)\Pr(BB)}$$

$$= \frac{(1/2)(1/3)}{(1)(1/3) + (1/2)(1/3) + (0)(1/3)} = \frac{1}{3}$$

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2. Random Variables

2.1 Motivation and Concept

In many experiments, interest lies in a **function of the outcome**, not the outcome itself.

Examples discussed in lecture:

- sum of two dice
- number of heads in coin tossing

A **random variable** is a real-valued function defined on the sample space.

Definition:
A random variable X on a sample space Ω is a function:

$$X : \Omega \to \mathrm{R}$$

Values are determined by outcomes of the experiment.
Probabilities are assigned to possible values.

2.2 Distribution of a Random Variable

Two important components:

1. the set of values the random variable can take
2. the probabilities with which it takes those values

For any value a:

$$\{\omega \in \Omega : X(\omega) = a\}$$

is an event, abbreviated as $X = a$.

The collection of probabilities:

$$\Pr(X = a)$$

for all possible values a is called the **distribution** of X.

2.3 Visualization

The distribution of a random variable can be visualized as a **bar diagram**:

- x-axis: values of the random variable
- height of bar at a: $\Pr[X = a]$

2.4 Example

Experiment: Tossing 3 fair coins
Let Y = number of heads.

Possible values: $0, 1, 2, 3$

$$\Pr(Y = 0) = \frac{1}{8}, \quad \Pr(Y = 1) = \frac{3}{8}, \quad \Pr(Y = 2) = \frac{3}{8}, \quad \Pr(Y = 3) = \frac{1}{8}$$

Since Y must take one of these values:

$$1 = \sum_{i=0}^{3} \Pr(Y = i)$$

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3. Probability Mass Function (PMF)

3.1 Concept

A random variable that can take **at most a countable number of possible values** is said to be **discrete**.

Let X be a discrete random variable with range:

$$R_X = \{x_1, x_2, x_3, \ldots\}$$

The function:

$$p(x) = \Pr(X = x)$$

is called the **Probability Mass Function (PMF)** of X.

Since X must take one of its possible values:

$$\sum_{x} p(x) = 1$$

3.2 Example

The PMF of a random variable X is given by:

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

where $\lambda > 0$.

Since:

$$\sum_{i=0}^{\infty} p(i) = 1$$

$$c\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Using:

$$e^{\lambda} = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$$

$$ce^{\lambda} = 1 \Rightarrow c = e^{-\lambda}$$

Thus:

$$\Pr(X = 0) = e^{-\lambda}$$

$$\Pr(X > 2) = 1 - \Pr(X \leqslant 2) = 1 - \Pr(X = 0) - \Pr(X = 1) - \Pr(X = 2)$$

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4. Class Participation

Students are instructed to switch to **Campuswire** for participation and discussion.

If you want, I can also:

- convert this into **LaTeX exam notes**,
- compress it into a **formula-only cheat sheet**, or

- format it exactly like a **handwritten lecture scribe**.